

Interplay of Security, Privacy and Usability in Videoconferencing

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ABSTRACT

In the period from August to October 2022, we conducted an online survey concerning users' perception and use of videoconferencing systems. We collected and analyzed a total of 245 responses from German-speaking participants. Each participant reported general use and case-specific use scenarios. We present usage trends and perceptions with respect to security, privacy and usability of various unified communications and collaboration (UCC) applications. The most frequently used tools, such as Zoom and Microsoft Teams, were rated lower in terms of security and privacy, but their usability ratings were above average. Tools tend to be avoided because of a lack of usability rather than security and privacy concerns. Nevertheless, almost 12% (29/245) of all participants reported using applications which were subject to security and privacy issues and which study participants would therefore personally prefer to avoid.

CCS CONCEPTS

• Security and privacy → Usability in security and privacy.

KEYWORDS

videoconferencing, unified communication, UCC, security, privacy, usability, online meetings

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1 INTRODUCTION

In spring 2020, the outbreak of the COVID-19 pandemic led to worldwide restrictions on face-to-face contacts [14]. Among other things, this caused a significant increase in usage of long-distance communication. Globally, the unified communications and collaboration (UCC) market stood at \$47 billion in 2020, growing 24.9% that year [18]. Microsoft Teams had 200 million daily active users in April of the following year (up 30 million since October 2020). Daily Zoom meeting participants were as high as 300 million at that time (2020: 200 million, end 2019: 10 million) [15].

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In connection with social distancing, daily work communication shifted to videoconferencing systems [2]. Private meetings with friends and family, leisure activities, sports and culture also took place via network-based interaction. At the same time, security incidents and intrusions into users' privacy increased. Countless videos of "embarrassing" incidents during a video call surfaced on the Internet. Corresponding tools came under close scrutiny from security experts and data protectionists. Inadequate privacy and security practices were exposed [11, 17]. German data protection and supervisory authorities issued recommendations for use [3, 7, 8]. In particular, they addressed the use of videoconferencing applications within companies. However, individual users were largely left to their own devices. In this context, the question arises for which purposes and in which context videoconferencing applications are used. Depending on the context, users also have different requirements in terms of security and privacy [9]. The discovery of security-related vulnerabilities in various tools has partially led to applications being improved and, in some cases, now also offering security and protection features [20]. Yet, it remains in question to what extent these measures influence user behavior and usability, and whether possible restrictions are accepted in favor of increased protection. The following research questions are investigated in this work:

- (1) In which context are videoconferencing tools used?
- (2) What is the interplay of security, privacy and usability of videoconferencing tools?

Of particular interest is the extent to which users reveal usage changes regarding comparative periods before the pandemic, during the pandemic and in the months after.

2 BACKGROUND AND RELATED WORK

Emami-Naeini et al. [9] call for a in-depth understanding of users' behaviors, attitudes, and concerns about aspects of remote communications. Such deep understanding is the only way to protect users and respect their privacy. To this end, Emami-Naeini et al. conducted a global online survey in May 2020 with 220 participants. Among the most frequently cited factors affecting users' comfort level were security and privacy. Likewise, the authors found that most users lack autonomy when it comes to deciding which tools and functions to use. In particular, camera/microphone activation has a negative impact on personal perceptions of privacy. While they studied two time periods (before and during the pandemic), we also consider the time after social distancing. Moreover, we limited the study to German-speaking participants, thus focusing on a specific population, as Emami-Naeini et al. had suggested for future work. Additionally, we not only examined videoconferencing from the home, but also on the road and at work usage.

Reisinger et al. [13] conducted a systematic security and privacy analysis of various videoconferencing tools. For this purpose, 10 well-known and widely used unified communications (UC) platforms were analyzed. Based on this analysis, we presented our users with 10 examples of threats that could occur in real life. An overview of the security and privacy features of these UC architectures led Reisinger et al. to the conclusion that, while confidentiality can be ensured by end-to-end encryption, other privacy features are often lacking. With this in mind, we also looked in particular at the non-technical measures users take to protect their privacy and communications. The German Federal Office for Information Security (BSI) [4] considered in particular the protection of videoconferencing systems in areas with increased protection requirements. Against this background, basic system components, technical requirements and practical challenges for the implementation of corresponding solutions are discussed. Their work has helped us to consolidate our definition of videoconferencing systems and thus to define tools relevant to our investigation as well as to identify important or possibly critical features when using these systems. To the best of our knowledge, our study is the first to directly investigate the interplay between security, privacy and usability.

3 METHODS

We conducted a German-language online survey between August and October 2022. Our approach was exploratory and descriptive. For the implementation of the questionnaire and for data collection we used LimeSurvey [12], a free and anonymous online survey tool (as on-premises solution in our institution).

3.1 Participant Recruitment and Demographics

Survey participants were recruited through a snowball method. Via Facebook, WhatsApp and e-mail, personal contacts were asked to participate and to forward the survey to their own contacts. All participants had used videoconferencing applications within the two years prior to the study. As an incentive for participation, a raffle for 15 Amazon gift cards worth €10 was offered. Of the original 255 full responses, 245 valid records were evaluated. Ten participants were excluded from evaluation: In one case, the questionnaire was paused several times and took over an hour to complete, two participants completed the survey in less than five minutes (too short to answer conscientiously), and three people added relevant tools manually (excluding further application-related questions). Four participants had inconsistent responses within individual usage scenarios. Detailed demographic data of the participants can be found in Table 1.

The included 245 participants are aged between 19 and 69 years. Most of them were employed (131) or students (97). There were slightly more male (130/245) than female (109) participants. Almost a third (79) had a professional background in computer science or in a similar field. 41 persons had not (yet) completed vocational education at the time of study, 38 participants reported an apprenticeship and 21 a specialised degree (e.g., master craftsman or technician) as their highest professional degree. 141 out of 245 had a bachelor's, master's, or doctoral degree. While one person was still attending school at the time of the survey, 206 out of 245 had a high school

diploma or comparable qualification. 36 participants reported a lower academic education.

3.2 Survey Procedure

In addition to eight questions on demographic data, the survey contains 47 content-related questions. Figure 2 in the appendix provides a simplified overview of the survey structure. The first group of questions deals with software use and limitations in choice. Used tools are described according to time period, frequency, purpose of use (professional/private/both) and usability rating. Subsequent questions relate to general experiences with functions, locations of participation, devices used, embarrassing/unpleasant situations and measures taken to protect privacy and communication. Furthermore, participation in training and education is queried. Likewise, experiences with security and privacy problems during videoconferencing, and the resulting reactions of participants, are enquired about. Tools are also rated in terms of security and privacy. The general use is concluded with an optional statement on events already attended or organized in the form of a video conference. Context-specific factors of use are addressed with an individual usage scenario (of the respondent's own choice). After specifying the physical and social context as well as the work equipment, the respondent is asked to evaluate his or her own usage scenario. The final set of questions identifies potential threat scenarios related to the general use of videoconferencing applications. Participants are asked to what extent they consider these threats to be realistic. Optionally, participants can also add which threats they fear in relation to their own conferencing activities.

3.3 Quantitative and Qualitative Data Analysis

The data analysis includes both the evaluation of quantitative and qualitative results. Most questions are based on fixed answer options and were thus analyzed quantitatively using R-Studio.

Open-ended answers were collected about individual usage scenarios. These questions were mandatory and thus received feedback from all participants. The data were inductively coded by the first author, subsequently discussed in several team meetings and adjusted. We identified twelve use cases. Table 2 provides an overview of observed usage scenarios and the number of people in the corresponding category.

3.4 Ethics

The questionnaire was approved by our university's data protection office. All participants gave their voluntary and informed consent to collection of data in the beginning of the survey. Data collection was completely anonymous and providing demographic information was voluntary. Participation in the raffle (with e-mail address) was implemented via a separate survey to ensure anonymity of the survey data. E-mail addresses were only used for the prize distribution and deleted afterwards.

4 RESULTS

We asked participants about three time periods: Usage before lockdown in spring 2020, usage during the first lockdown, and during the three months prior to our study. Results showed that social distancing had and still has a significant impact on usage.

Table 1: Participants' demographic information, N = 245.

Age		Gender		Education (professional)		Employment		IT Background	
19-29	64%	Female	45%	No vocational education	17%	Employed	54%	Yes	32 %
30-49	31%	Male	53%	Vocational education	16%	Student	40%	No	66%
50-64	3%	NA	2%	Specialised degree	9%	Pupil/Apprentice	2%	NA	2%
> 64	1%			Bachelor's degree	28%	Self-employed	2%		
NA	2%			Master's degree	26%	Home-maker	2%		
				Doctorate degree	4%	Retired	2%		
				NA	2%	Unemployed	1%		
						NA	2%		

4.1 Periods and Frequency of Use

Over a third of all participants (85/245) had never used videoconferencing before the lockdown in spring 2020. Within the last three months before the study, almost 90% (220/245) of participants continued to use UC tools. Figure 1 compares the absolute number of users per application for the examined periods. Further information on use frequency per tool for each time period can be found in the appendix in Figure 3.

4.1.1 Before the Pandemic: Of those participants who had experience with videoconferencing before the pandemic, 72% (115/160) used Skype. Only 12% had already used Skype for Business or Microsoft Teams at this point. Discord (n = 45) was used slightly more than Zoom (n = 40) at this time. Before the pandemic, the tools were also used less than once per month in over 60 percent of cases.

4.1.2 During the Pandemic: Contact restrictions also shifted the distribution of the most frequently used tools. Within the two years in which contact restrictions prevail, Zoom (85%, 209/245) and Microsoft Teams (74%, 181/245) were used extensively. Skype, the only tool that has not seen an increase in the number of users during the pandemic, was being used by nearly as many users as before the contact restrictions (n = 114). Tools used by the majority of their respective users more than monthly were Skype for Business (31/58, 53%), Zoom (127/209, 61%), and Microsoft Teams (112/181, 62%). Others were mostly used monthly or less frequently.

4.1.3 After the Pandemic (three months prior to the study): We observed an overall decrease in usage. Most users continued to use Microsoft Teams on a weekly or even daily basis. However, the number of Teams users overall dropped by 23%, and Zoom decreased by 35%. Here, too, monthly or even less frequent use within this time period was now more frequent (52%). Compared to other applications and especially compared to the period before the pandemic, the usage figures for both tools remained high (Zoom: 155/245, MS Teams: 139/245). In contrast, only 45 from original 115 participants continued using Skype.

4.2 Context of Use

The majority (202/245) of participants attended video conferences for three to eight purposes. Only 13% limited their use to one or two use cases. The remaining 5% were involved in more than eight types of events. The appendix contains an overview of the intended purposes of video meetings that were organized or attended by

participants (see Figure 4). With regard to case-specific scenarios chosen by participants, twelve use cases can be distinguished. These can be summarized in three areas: Education and teaching (n = 85), professional communication and advice (n = 101) as well as leisure activities and interest groups (n = 59). A more detailed presentation of use cases, taking into account the role of the respondent and most frequently used applications, can be found in Table 2.

4.2.1 Education and Teaching: Zoom is used in particular in the field of education and teaching. Overall, it was used in 64% (54/85) of these use cases. However, a more specific distinction revealed that formal education was mostly provided via Microsoft Teams. Education and teaching was also the only area in which BigBlueButton was used, albeit to a small percentage (5/85).

4.2.2 Professional Communication and Advice: In all areas of professional exchange, Microsoft Teams (55%, 56/101) was used most frequently. Video conferences concerning company-internal cooperation (team meetings), as well as the area of cross-company cooperation (collaboration), used Microsoft Teams in more than half of the cases. Zoom is only used in 16 out of 101 cases in the professional environment, followed by Skype for Business (n = 7) and Webex (n = 5).

4.2.3 Leisure Activities and Interest Groups: Video conferences related to leisure activities also used Zoom most frequently (36%, 12/59). Discord, which was used in a total of 20 percent of these cases, also accounted for a large part of leisure activities. The tool was used when participants met with friends and family, especially during video calls with acquaintances to play video games (online gaming).

4.3 Dealing with Privacy and Security Issues

26% of all participants had never heard of problems with security and privacy of videoconferencing applications. Conversely, almost three quarters had already been aware of such incidents (e.g., through media reports). For almost half of them (87/181), however, this had no influence on their use.

Responses to known Privacy and Security Issues: The remaining 94 respondents described different reactions to corresponding problems: 42 people subsequently used tools they consider less critical, 39 became more cautious about using videoconferencing tools in general, and 34 had adjusted the security settings of the affected application. However, there were also 29 people who continued to

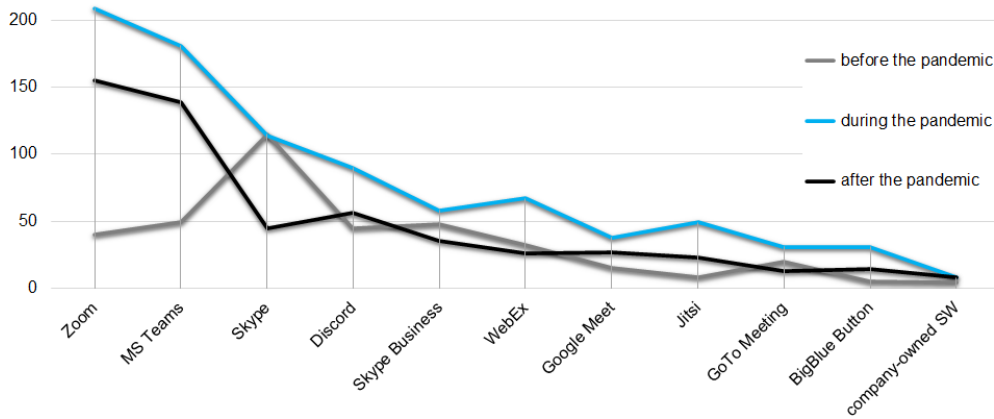


Figure 1: Absolute number of users per tool over three time periods, max. = 209, N = 245.

Table 2: Classification of the described usage scenarios considering the role of the participant and indication of the most frequently used tool. Each of 245 participants described one scenario, n = number of scenarios per use case.

Use Cases (N = 245)	n	organizing/ moderating	presenting	interactive/ passive	Most frequently used application
Education and Teaching (n = 85)					
Formal education	14	36%	21%	43%	MS Teams 64%
Higher education	43	12%	28%	6%	Zoom 79%
Further training	15	20%	20%	60%	Zoom 47%
Private lessons	6	67%	-	33%	Zoom 70%
Study groups	7	29%	29%	43%	Zoom 90%
Prof. Communication and Advice (n = 101)					
Job interviews	10	-	40%	60%	MS Teams 50%
Team meetings (internal)	58	34%	22%	43%	MS Teams 57%
Collaboration (external)	21	76%	10%	14%	MS Teams 62%
Customer services	12	75%	-	25%	MS Teams 42%
Leisure Activities and Interest Groups (n = 59)					
Friends and family	31	52%	10%	39%	Zoom 32%
Sports/culture/associations	21	52%	-	48%	Zoom 52%
Online gaming	7	29%	-	71%	Discord 86%

use such applications even though they would have rather avoided them.

In this context, P221 explicitly reported a lack of decision-making regarding the choice of application: "You effectively have no choice, either give [a] lecture and accept or don't give [a] lecture". Another participant (P52) limited the use of questionable applications to those that had to be used for "operational purposes/reasons".

Limited Decision-Making Autonomy: With regard to individual functions, almost 30 percent of the participants (70/245) addressed functions that contradict their personal privacy preferences, but were necessary to achieve the tasks and goals of the video call in a specific use case. Which functions were affected in different use cases can be looked up in Table 3. It also shows, among other things, that 54 people who used a camera in their described usage scenario would actually have preferred to do without it. In these cases, the camera was necessary to achieve the underlying tasks and goals

of the intended use. This particularly affects communications in the work environment (e.g., job interviews: 30%, team meetings: 28%, collaborations: 29%), but participants would also prefer to do without camera participation in three out of six one-on-one conversations between teachers and students. In online conversations with friends and family, on the other hand, only one person (1/31, 3%) wanted to do without this function.

4.4 Security and Privacy vs. Usage and Usability

We asked survey participants to rate used applications in terms of usability and security/privacy using a five-point Likert scale. In addition, participants were asked whether they had already chosen or avoided certain tools themselves. While any tools could be mentioned in the case of avoidance for security/privacy reasons, avoidance for a lack of usability only includes those applications

Table 3: Proportion of participants using functions (necessary for goals and tasks) that contradict their security and privacy preferences in the described use scenarios, n = number of use scenarios per use case, N = 245.

Use Cases	n	None	Audio	Video	Chat	Reactions	Screen Sharing	Filter
total in %	100%	70%	6%	22%	2%	2%	2%	2%
total in n	245	171	15	54	6	6	5	6
Formal education	14	71%	-	21%	-	-	-	7%
Higher education	43	63%	12%	26%	-	2%	2%	2%
Further training	15	67%	7%	20%	-	20%	-	13%
Private lessons	6	50%	-	50%	-	-	-	-
Study groups	7	71%	-	29%	-	-	-	-
Job interviews	10	70%	10%	30%	-	-	-	-
Team meetings	58	69%	9%	28%	5%	2%	5%	3%
Collaboration	21	52%	5%	29%	10%	-	5%	-
Customer services	12	67%	8%	25%	-	-	-	-
Friends and family	31	94%	-	3%	3%	-	-	-
Sports/culture/associations	21	76%	5%	10%	-	5%	-	-
Online gaming	7	86%	-	14%	-	-	-	-

Table 4: Comparison of average evaluation results in terms of user-usability and security/privacy (very good = 1, good = 2, neither good nor bad = 3, bad = 4, very bad = 5), n = number of ratings, $\emptyset$ = mean of all ratings, discrepancy = $\emptyset$ usability - $\emptyset$ security/privacy.

Applications	Users in n	Usability $\emptyset$	Security/Privacy $\emptyset$	Discrepancy $\emptyset - \emptyset$	% of Avoidance due to perceived lack in	
					Usability (N=n)	Security/Privacy (N=245)
Zoom	216	1,91	2,76	- 0,85	8%	11%
MS Teams	200	1,94	2,42	- 0,48	16%	4%
Discord	92	2,11	2,67	- 0,56	17%	3%
Firmen SW	11	2,45	1,55	0,90	18%	-
Skype Business	77	2,47	2,53	- 0,06	18%	2%
Google Meet	51	2,53	2,78	- 0,25	6%	5%
BigBlueButton	38	2,55	2,08	0,47	26%	-
Jitsi	54	2,61	2,22	0,39	20%	1%
Skype	155	2,64	2,91	- 0,27	31%	6%
WebEx	82	2,71	2,41	0,30	28%	2%
GoToMeeting	43	2,79	2,81	- 0,02	26%	1%

that have been used at least once. Table 4 shows the overall average of both evaluations (n lists all persons who used the tool and therefore also rated it), the discrepancy between them, and the percentage of users who had already given up software due to deficiencies in usability or security/privacy.

Discrepancy in Perception of Security/Privacy and Usability: In most cases there were discrepancies within both ratings of the same software. Although Zoom, MS Teams and Skype were some of the lowest rated tools in terms of security/privacy, they are also the applications most frequently chosen by participants themselves (Zoom: 116/216, Teams: 104/200, Skype: 61/155). However, more than 30 percent gave up Skype due to a lack of usability. Zoom and MS Teams, on the other hand, are the applications that are rated best for their usability. In total 20% (n = 50) of all participants had already avoided certain videoconferencing tools due to security/privacy concerns. On the other hand, more than twice as many users (49%) gave up applications due to a lack of usability.

Controllability of the System: Closely related to the usability of a system is its controllability. We looked at several functions that can lead to unwanted situations or privacy intrusions through incorrect interaction. 59% of all respondents said they had experienced at least one incident in which one of these functions was accidentally (de)activated by themselves or other meeting participants. Audio (n = 97), screen sharing (n = 62) and reactions (n = 53) are mentioned most often. 77% of people who experienced such an incident subsequently found it unpleasant or embarrassing.

Education and Training: User education and training promote the safe handling of an application and thus also its controllability. Almost three quarters of all participants (181/245) stated that they had never been involved in training for videoconferencing tools. However, the majority (70%) of these persons were already responsible for hosting a meeting. P51 highlighted how incorrect handling can affect everyone involved in the meeting by reporting: "Host unmuted all participants (100)". The incident was accompanied

by an embarrassing and uncomfortable feeling. When asked what users felt personally threatened by, P89 also addressed this factor: *"If the host of the meeting is technically clumsy and therefore does not configure the conference securely"*.

Perception of Security and Privacy Threats: Various exemplary threats were presented to the participants for evaluation. Using a five-point Likert scale (very realistic = 1 - not at all realistic = 5), it was discussed which attacks were classified as realistic. Table 5 shows how many people assigned each threat to each value, as well as the overall average of all ratings. In general, survey participants tended to perceive threats from within the conference as more realistic. Non-authorized recordings by other conference participants were rated as rather realistic or even very realistic by 83%. In comparison, the recording of content by providers (68%) or authorities (38%) was classified as realistic in fewer cases. Incidents related to a privilege escalation by outsiders were classified by the majority (52%) of the participants as not corresponding to reality. This was also reflected in the measures users take to protect themselves. Protective measures for communication consisted in particular of (de)activating the microphone and camera if necessary. With regard to the camera, 90% (221/245) of the participants had already used these means to protect their privacy. When it comes to audio, there were even 93%. On the other hand, measures such as pseudonymization (20%) or end-to-end encryption (15%) were rarely used in general. Taking into account concrete usage scenarios, only 9% (22/245) of all participants used the latter.

5 DISCUSSION

Usability over Security/Privacy: According to our findings, Skype (45/115, decreased by over 60%) was the only tool whose usage declined rapidly in the wake of the COVID-19 pandemic. In contrast, applications like Zoom and Microsoft Teams were more in demand than ever. Skype not only lags behind other applications in its feature set, it is also the software that was most often abandoned due to lack of usability (48/115, 31%). In contrast to usability (121/245), security/privacy concerns (50/245) played a far smaller role in forgoing a tool. According to Reisinger et al. [13], in commonly used applications, a lack of technical measures with regard to privacy are particularly noticeable, while the security of such tools is relatively well protected. Likewise, software that focuses on data protection-friendly handling was less popular with our users. For example, BigBlueButton (38/245) and Jitsi (54/245), used only by a small percentage of participants, was rated better than many other tools from a security/privacy perspective, but evaluated as below average in terms of usability.

Security and Privacy Measures. A total of 209 people and thus 85% of all participants generally described the use of technical and non-technical measures to protect their privacy and communication in a specific usage scenario. These usually consisted of the need-based (de)activation of audio and video. However, it must be considered problematic that deactivating the microphone can prove to be ineffective, at least in relation to the provider. Yang et al. [19] found that the microphone input is monitored at regular intervals during the muting of common videoconferencing applications. One

of the few technical measures that protects both security and privacy is end-to-end encryption. In fact, this was only activated by a total of 22 participants within a scenario. Mostly it was used in private communication with friends (5/31) or in an educational environment (9/85). In scenarios of professional interaction, the use of this method by our participants was hardly ever described. A possible reason could be restrictions of functionality that come with it [10]. On the other hand, the users surveyed perceived threats from within the conference in particular. Unauthorized recordings during a meeting by other participants, the misuse of legitimate recordings or the sharing of inappropriate content, which has become known as zoom bombing [11], were classified as quite realistic. Fake identities within a conference [1] were also perceived as more realistic than, for example, access from websites or applications to the microphone/camera [19] or surveillance by authorities (see Table 5).

Tasks and Goals vs Security and Privacy Preferences. Emami-Naeini et al. [9] pointed out that most of their participants do not have autonomy in the choice of videoconferencing applications or the use of microphone/camera. In several cases, this conflicts with personal privacy and security preferences. Also in this study, about 12% of all participants described using a tool they would prefer to avoid for security and privacy reasons. A total of 70 people addressed one or even up to four functions in various usage scenarios that contradicted their personal attitudes toward security and privacy, but were nevertheless necessary for the underlying goals and tasks (see Table 3). While the study mentioned [9] was limited to the audio and video functions, we additionally critically examined other functions in terms of security and privacy. For example, when sharing screens, information that should not be visible to others or is even confidential may be inadvertently shared [4]. More than a quarter of participants (62/245) had also experienced an incident in which information was shared accidentally via the screen.

5.1 Limitations

Compared to the other two periods, only a small time span was examined by looking at the last three months before the study. Data protection, security and privacy are addressed directly by a number of questions. For this reason, priming with regard to these factors cannot be completely ruled out. Almost one third of the survey participants also came from a computer science background or have a similar professional background and may therefore be particularly sensitive to the topic. Data collected on the basis of a snowball system is not representative of the German population. In the 19-29 age group in particular, participation in the survey was above average (64%) [16]. The level of education was above the German average with 84% (cf. 35.5% [5]) of the participants who had high school diplomas or comparable academic degrees.

6 CONCLUSION

Many users were confronted with videoconferencing for the first time [6]. In our case, more than a third of participants used these tools for the first time due to the contact restrictions. After the pandemic, videoconferencing tools continue to play an important role in everyday life, especially in work contexts [2]. In our study only 64 out of 245 participants attended education and training on

Table 5: Potential threats on videoconferencing applications (based on x) rated by participants according to realism (by 5-point Likert scale from very realistic to not at all realistic, N = 245, % = proportion of N, Ø in total = mean of all ratings).

Threats	very realistic	rather realistic	neither nor realistic	rather not realistic	not at all realistic	Ø in total
Recording by persons involved	47%	36%	9%	7%	2%	1,84
Fake invitations for conference	30%	44%	12%	11%	4%	2,18
Recording by providers	28%	40%	17%	12%	3%	2,25
Abuse of legitimate records	22%	36%	21%	18%	3%	2,45
Sharing inappropriate content	18%	35%	13%	24%	10%	2,75
Fake identities in conference	18%	35%	13%	24%	10%	2,75
Access to camera/microphone	15%	41%	17%	24%	4%	2,63
Recording by authorities	14%	24%	22%	34%	6%	2,97
Access to devices used	8%	27%	24%	33%	9%	3,29
Admin privileges for conference	7%	24%	17%	40%	12%	3,09

tools (14 of them even on their own initiative). Both the BSI [4] and Reisinger et al. [13] emphasize the importance of education and training for secure handling. Of those 181 participants who did not attend training and education 126 were already responsible for (co)hosting a meeting. This results in an increased risk potential for everyone involved in the meeting. At the same time, employers do not take responsibility for providing sufficient guidance to employees. We therefore find it necessary to support users in handling tools and to develop appropriate programs for awareness, education and training. Above all, companies and also educational institutions should be encouraged to do so, especially when they require the use of certain tools and thus potentially also restrict personal preferences with regard to data privacy and security.

The data collected as part of the user study will be subjected to logistic regression in the further course of the study. In this way, the factors that significantly influence the usage of these tools will be identified. Furthermore, the preferences determined with regard to the use of protective measures and the omission of functions may also provide useful information for the further development of various applications on the part of providers, manufacturers and developers. Open source tools in particular seem to lag behind commercial applications in terms of usability, but in return they not only offer insights into the source code, but also provide a more privacy-friendly approach to users' personal data.

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A ADDITIONAL FIGURES

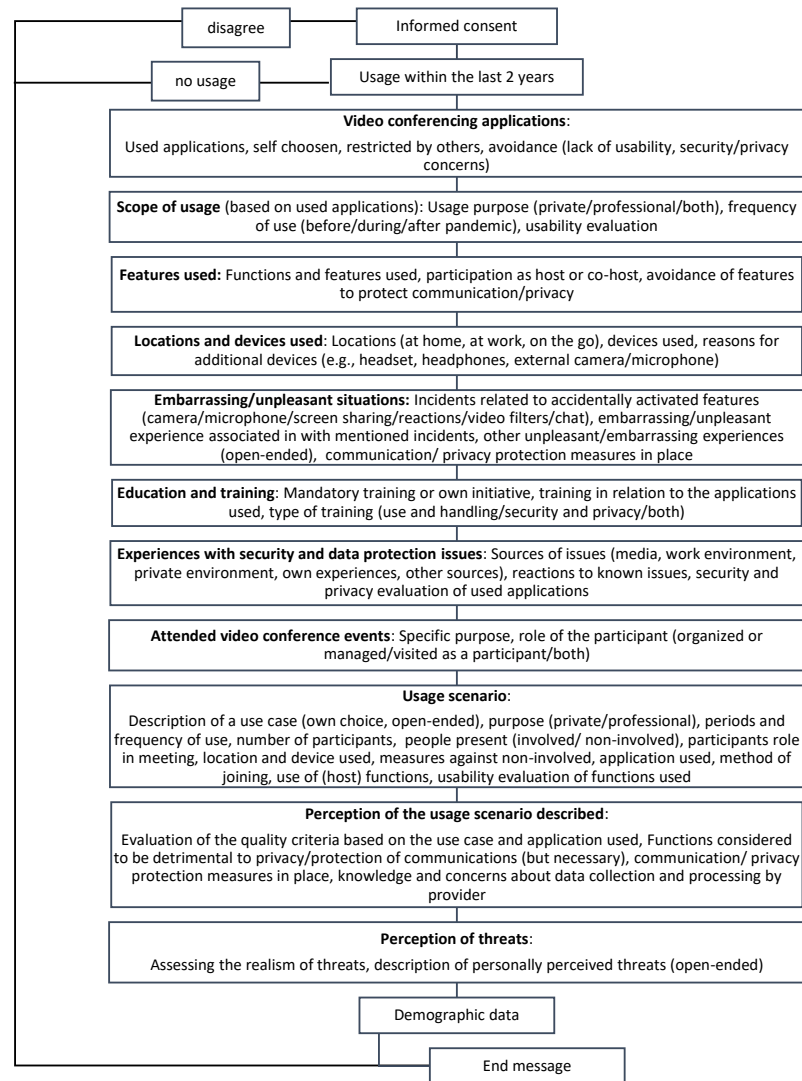


Figure 2: Simplified overview of the survey structure divided into question groups.

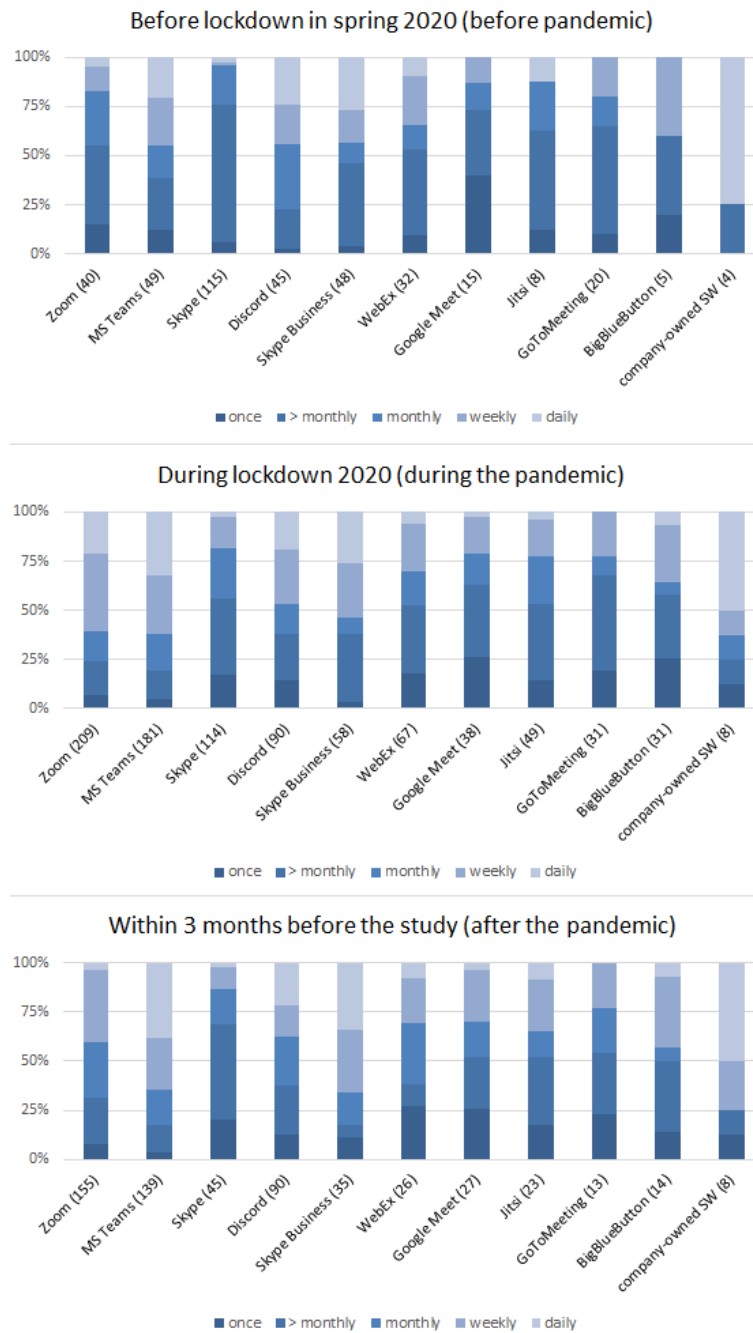


Figure 3: Proportion of participants in terms of reported frequency of use over 3 time periods: before pandemic, during pandemic and after pandemic, (n) = absolute number of users periods shown.

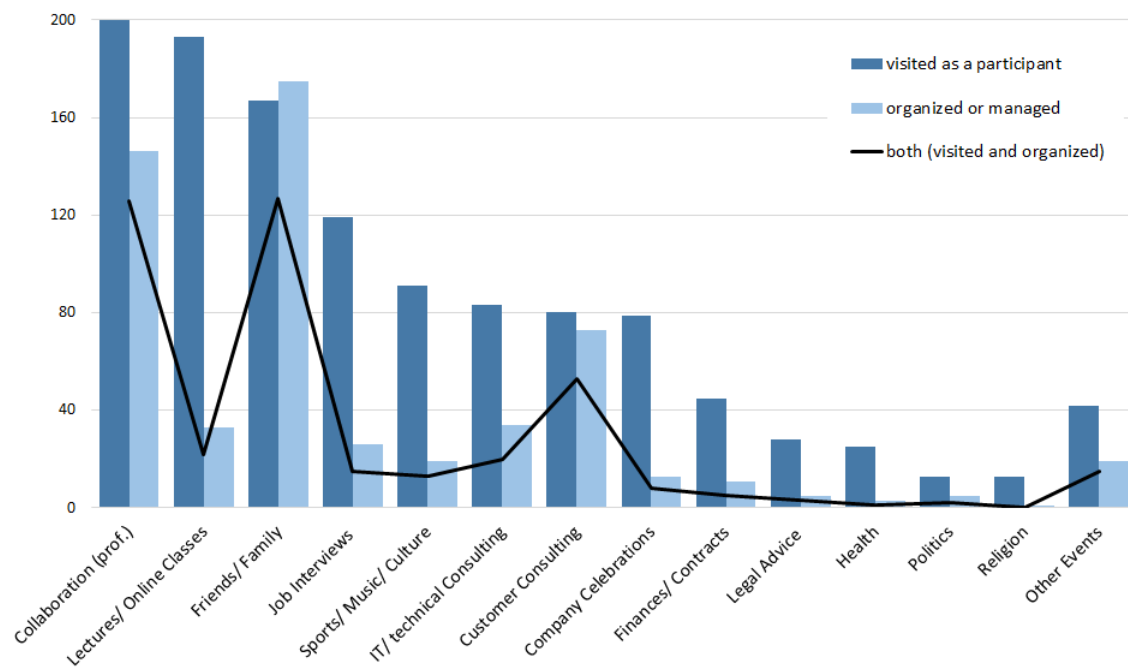


Figure 4: Video conferences attended or organized by participants, by number of people who were already part of the specified events, organized = guiding function as (co-)host/moderator/presenter, N = 245.